



Outline

Anatomy of a NN

Design choices

- Activation function
- Loss function
- Output units
- Architecture



Activation function

$$h = f(W^T X + b)$$

The activation function should:

- **Provide non-linearity**
- Ensure gradients remain large through hidden unit

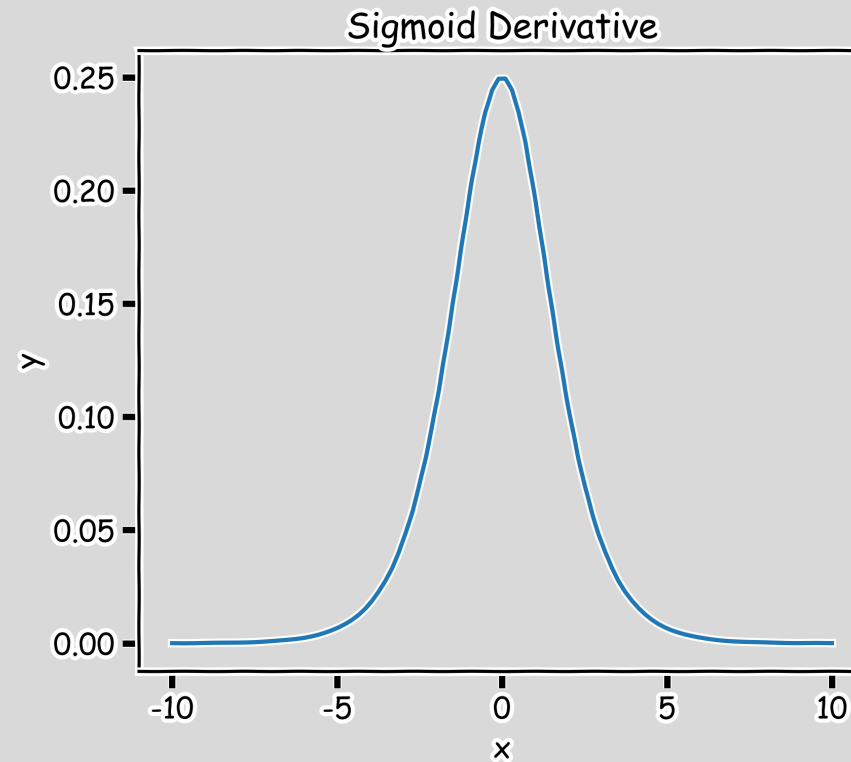
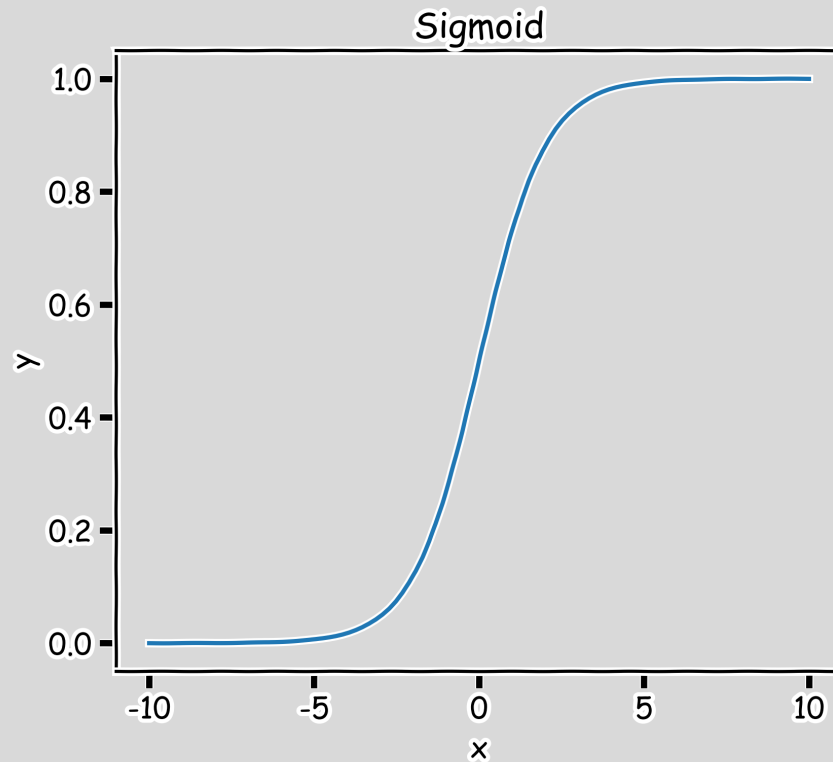
Common choices are

- Sigmoid
- Relu, leaky ReLU, Generalized ReLU, MaxOut
- softplus
- tanh
- swish



Sigmoid (aka Logistic)

$$y = \frac{1}{1 + e^{-x}}$$

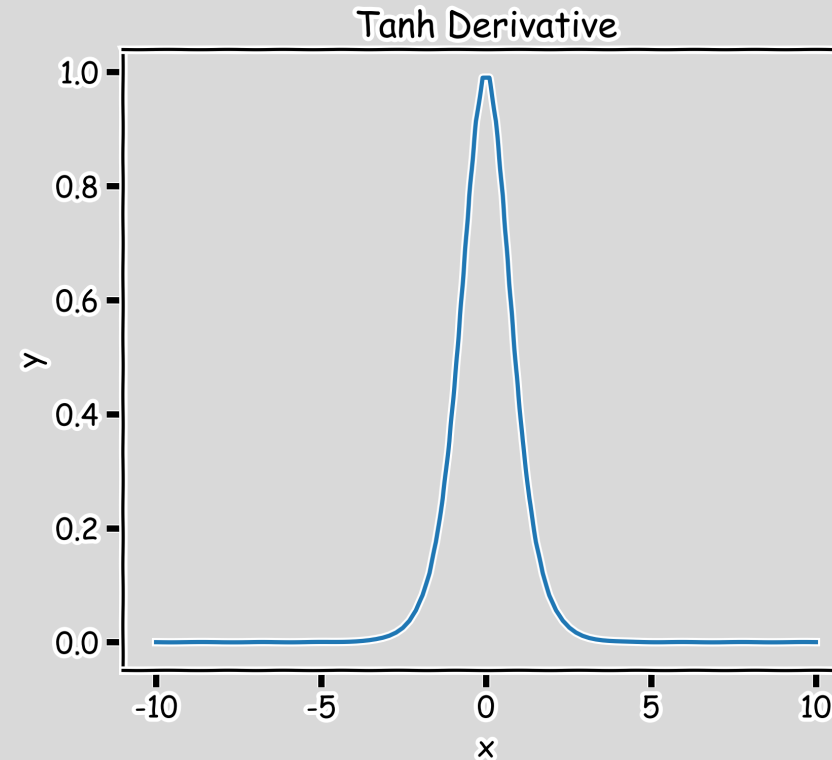
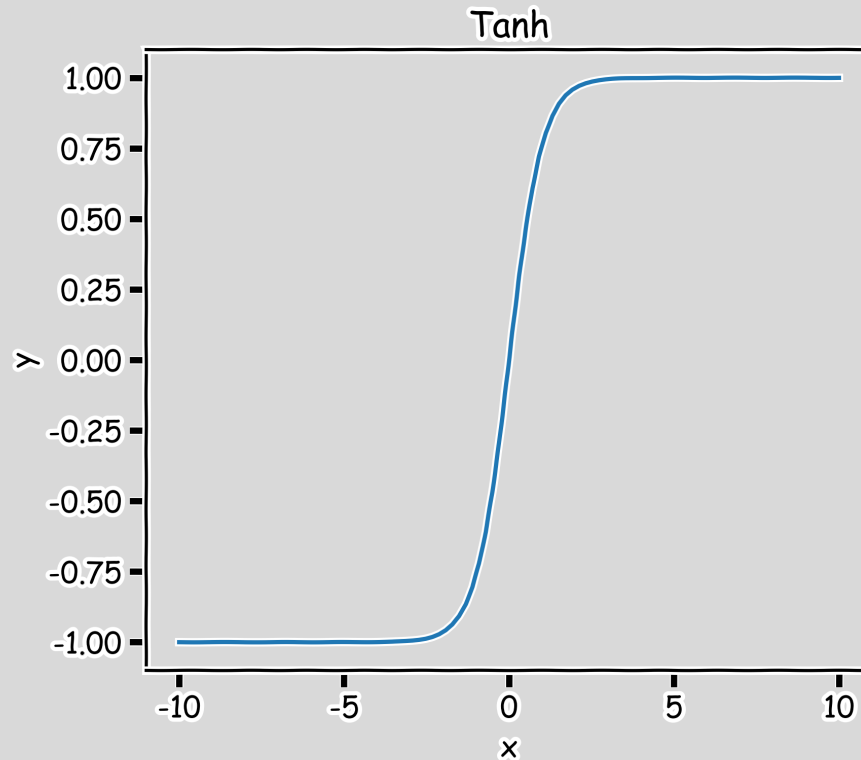


Derivative is **zero** for much of the domain. This leads to “vanishing gradients” in backpropagation.



Hyperbolic Tangent (Tanh)

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

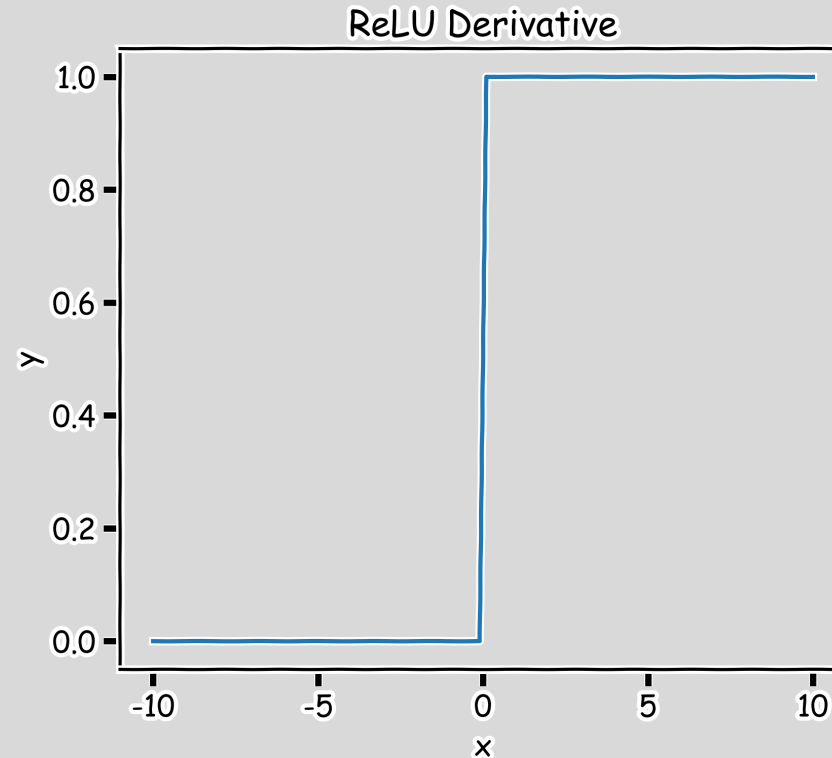
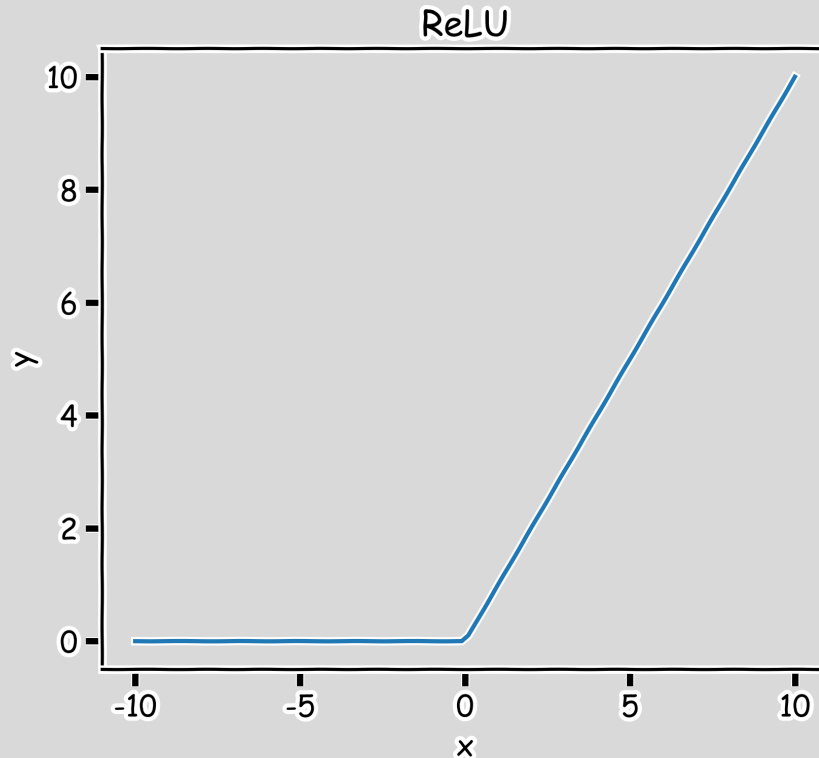


Same problem of “vanishing gradients” as sigmoid.



Rectified Linear Unit (ReLU)

$$y = \max(0, x)$$



Two major advantages:

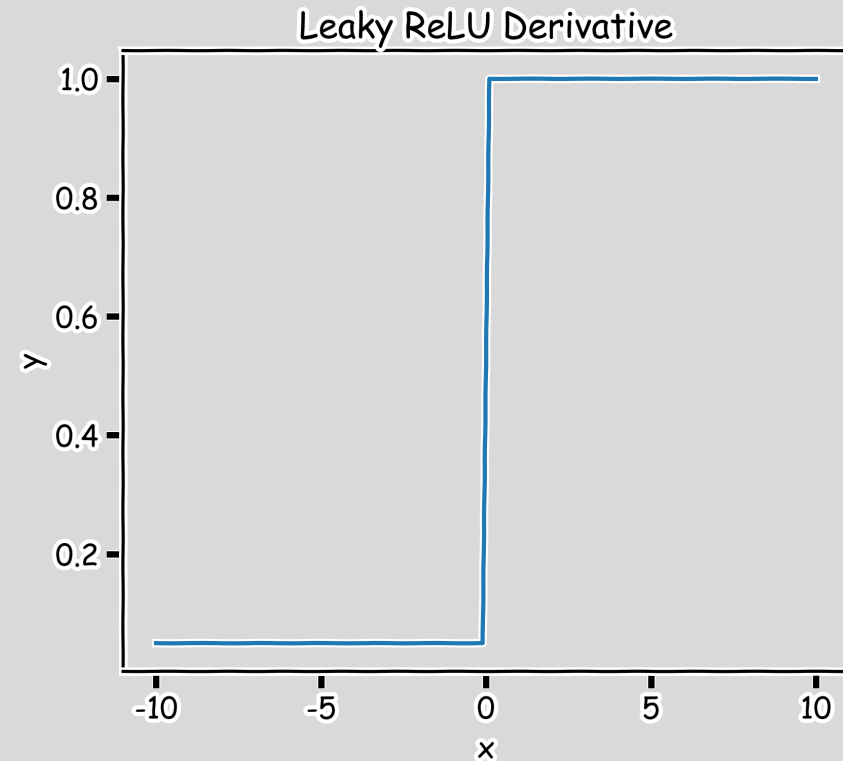
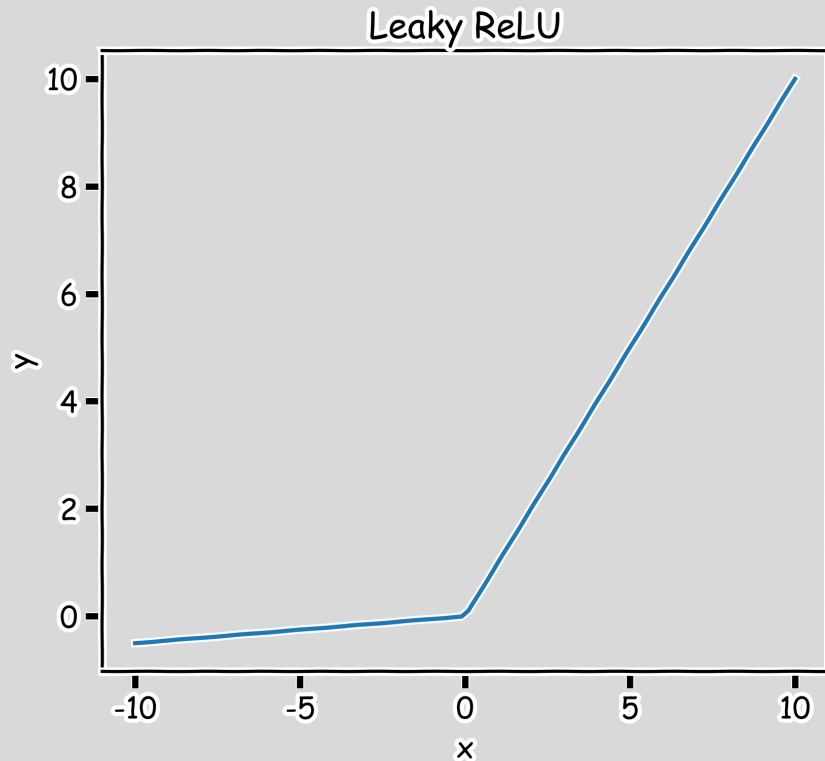
1. No vanishing gradient when $x > 0$
2. Provides sparsity (regularization) since $y = 0$ when $x < 0$



Leaky ReLU

$$y = \max(0, x) + \alpha \min(0, x)$$

where α takes a small value



- Tries to fix “dying ReLU” problem: derivative is non-zero everywhere.
- Some people report success with this form of activation function, but the results are not always consistent

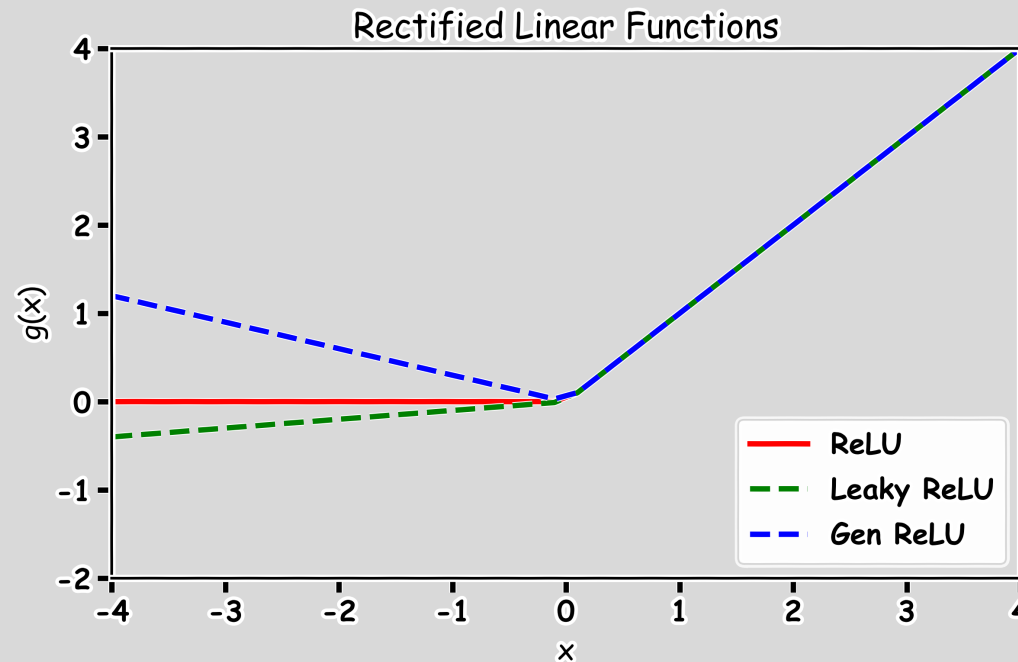




Generalized ReLU

Generalization: For $\alpha_i > 0$

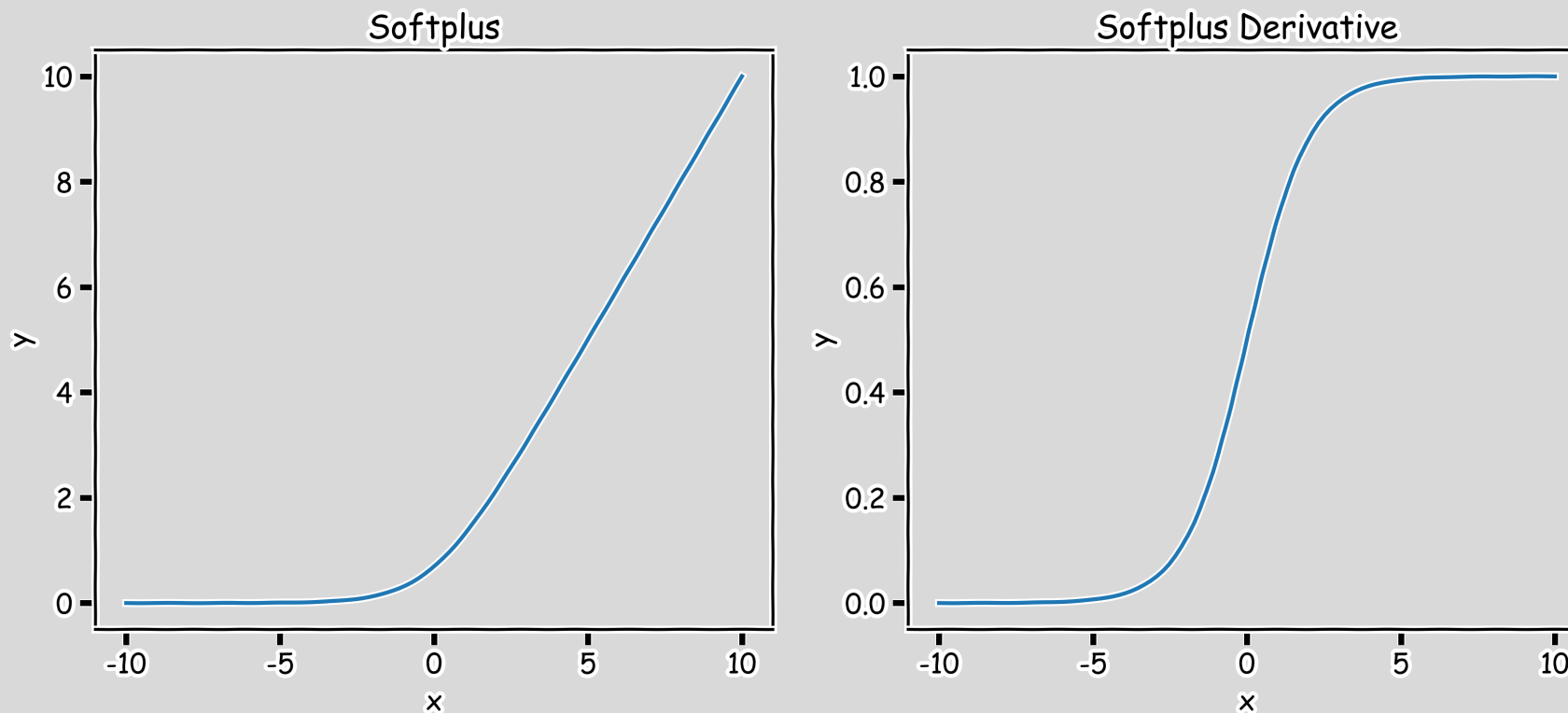
$$g(x_i, \alpha) = \max\{a, x_i\} + \alpha \min\{0, x_i\}$$





softplus

$$y = \log(1 + e^x)$$



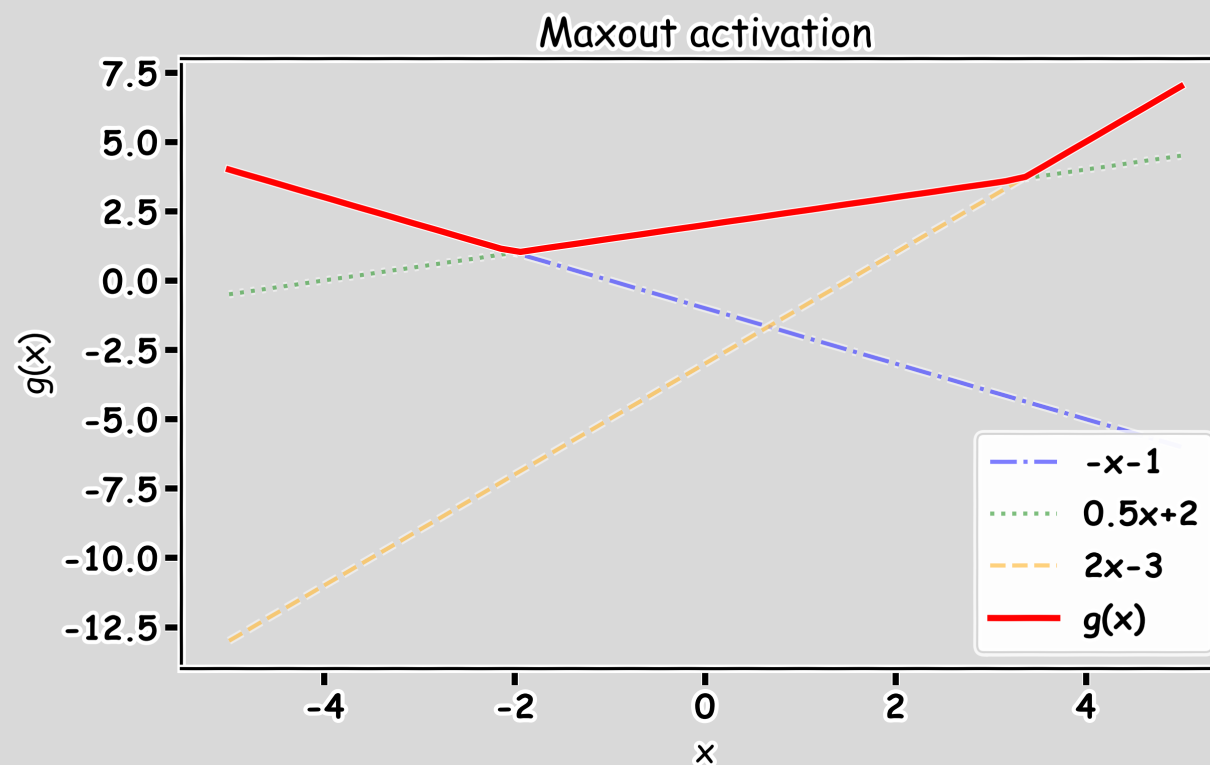
The logistic sigmoid function is a smooth approximation of the derivative of the rectifier



Maxout

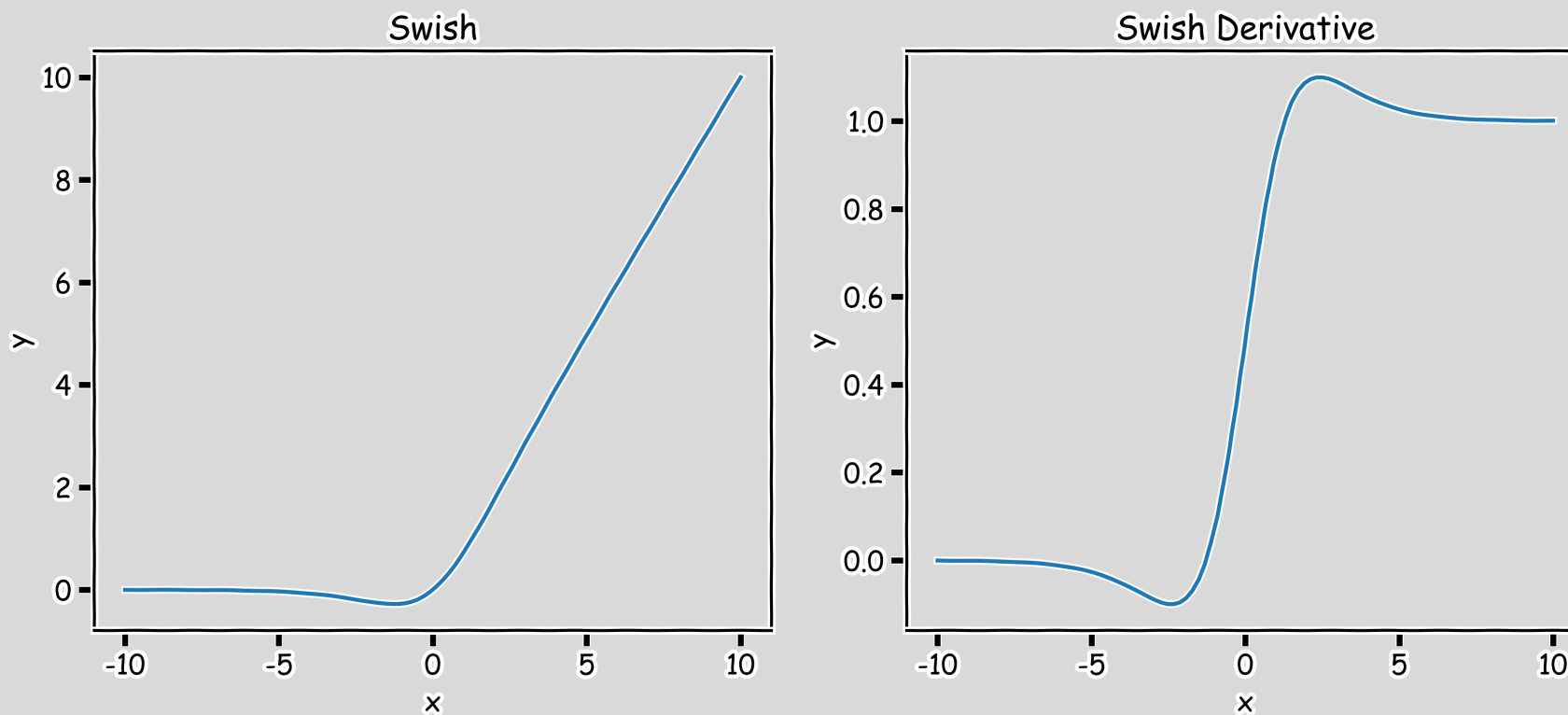
Max of k linear functions. Directly learn the activation function.

$$g(x) = \max_{i \in \{1, \dots, k\}} \alpha_i x_i + \beta$$





Swish: A Self-Gated Activation Function



$$g(x) = x \sigma(x)$$

Currently, the most successful and widely-used activation function is the ReLU. Swish tends to work better than ReLU on deeper models across a number of challenging datasets.

